COMP661 Deep Learning

Semester 2 2026

Assignment 2 - Kahikatea Tree Classification Using Deep Learning

Worth **: 35%**

Submission **: URL for the GitHub Repository**

Dataset **: kahikatea.zip (Download from the link given in Dataset Description)**

Final Submission Due: Friday, 18 September 2026 5:00 p.m.

Late Penalty: Work submitted after the due date and time attracts a maximum late penalty, as noted below, of the marks available:

Less than 6 hours late	5% late penalty
Between 6 and 24 hours late (i.e., up to 1 day)	12.5% late penalty
Between 24 and 48 hours late (i.e., 1 - 2 days)	25% late penalty
Over 48 hours late (i.e., 2 days or longer)	100% late penalty

No work will be accepted after Sunday, 20 September 2026 5:00 p.m.

This is an **individual** assessment. You must not collaborate or confer with others. You may help others by verbally explaining concepts and making suggestions in general terms, but without directly showing or sharing your own work. You must develop your code, visualisations, and interpretations independently on your own, even if working alongside others. Any work that is copied or contains identical code/visualisations/interpretations to others will receive zero marks for both parties involved.

The use of Artificial Intelligence (AI) tools, such as ChatGPT or Microsoft Copilot, to complete this assessment is **prohibited**. Assessment answers will be analysed for evidence of the use of AI and penalties may be administered.

The University policy on Academic Integrity can be found here.

Background:

Accurate mapping of native vegetation is essential for biodiversity conservation, ecosystem restoration, and sustainable land management in New Zealand. Traditionally, identifying tree species from aerial photographs requires significant time and expert knowledge. Recent advances in deep learning and computer vision have made it possible to automatically identify vegetation types from high-resolution aerial imagery, supporting conservation agencies in monitoring native forests over large geographical areas.

In this assessment, you will investigate the **Kahikatea Image Dataset**, which was developed as part of the **TAIAO Environmental Data Science** (Jia, 2021). The dataset contains aerial images extracted from orthophotographs captured over New Zealand landscapes. Each image has been annotated to indicate whether it contains Kahikatea (*Dacrycarpus dacrydioides*), one of New Zealand's tallest native tree species, which commonly grows in lowland swamp forests.

For this assessment, your primary objective is to develop and evaluate deep learning models with different configurations and architectures for binary image classification to determine whether a given aerial image contains Kahikatea.

You are expected to investigate the dataset, perform data preparation and exploratory analysis, design and evaluate deep learning models, and critically assess the strengths and limitations of your approach.

Dataset Description:

Each image in the Kahikatea dataset represents a small geographic area and has been labelled based on the presence or absence of Kahikatea trees.

You can download the dataset with this link:

https://datasets.cms.waikato.ac.nz/taiao/kahikatea/

Folder Structure:

The downloaded dataset is organised into the following folder structure.

```
kahikatea/
 ├ expl/
     ├ ve_positive
        └ Biospatial_20160702U_1861_01_03.png
        └ Biospatial_20160702U_1861_01_04.png
       ...
     ├ ve_negative
 ├ data/
     ├ve_positive
        └ Biospatial_20160702U_2386_01_03.png
        └ Biospatial_20160702U_2386_01_04.png
       ...
     ├ ve_negative
        └ Biospatial_20160603U_1502_01_01.png
        └ Biospatial_20160603U_619_02_02.png
       ....
```

The **data** folder contains the image files which are organised into sub-folders corresponding to image classes:

- **ve_positive:** aerial images that contain Kahikatea trees (Dacrycarpus dacrydioides).
- **ve_negative:** aerial images that do not contain Kahikatea trees.

In the dataset naming convention, **ve** refers to the vegetation image category. The dataset also includes a separate **expl** folder containing segmentation masks that indicate the locations of Kahikatea trees within positive images. These masks are intended for advanced tasks such as object localisation and image segmentation and are available only for images in the **ve_positive** class. **You do not need to use the expl folder for this assessment and may ignore it.**

The dataset consists of 232 positive images and 287 negative images.

The Kahikatea dataset presents several real-world challenges, such as variations in lighting and shadows and differences in tree size, age and density which make image classification more difficult.

You are encouraged to explore the dataset before developing your models and use your observations to guide your modelling decisions.

Instructions:

Setting up your repository

Create a private GitHub repository called **StudentID_Kahikatea_Project2** (replace "StudentID" with your actual student ID).

Invite **lincolnmac (computing@lincoln.ac.nz)** as a collaborator with read access so your submission can be accessed for marking.

Follow the folder structure given below.

```
123456789_ Kahikatea_Project2/
 ├ README.md
 ├ requirements.txt
 ├ data/
     └ (you do not have to copy the images here. Instead, please provide instructions to download the dataset in the README.md file.)
 ├ src/
     └ EDA.ipynb
     └ DL.ipynb
 ├ figures/
     └ (saved visualisations, e.g. target_distribution.png)
 ├ models/
     └ final trained model files
```

Commit History

Your repository must show at least two meaningful commits per week, from the day you create the repository right through to the final submission deadline. This standard applies across the entire project. Your commit history forms part of the assessment.

Notebook Structure

The project is built around two Jupyter notebooks, both located in the src/ folder.

1. **EDA.ipynb** contains the exploratory data analysis and preprocessing (answers for Task 1 and Task 2).
2. **DL.ipynb** contains the modelling work and discussion (answers for Task 3 and Task 4).

The dataset should be loaded using relative paths from within **src/**. Both notebooks must execute successfully from start to finish in a clean environment, using only the packages listed in **requirements.txt**. Each code block should use meaningful variable names and appropriate comments. Precede each code block with a Markdown cell explaining its purpose and follow it with a Markdown cell interpreting the results.

Each notebook must contain clear section headings and present figures and tables with appropriate titles and captions. All discussions and reflections should be added to markdown cells of corresponding notebooks.

Any preprocessing or modelling decision should be justified with reference to patterns or evidence observed in the data.

The '**data**' folder must be added to **.gitignore** to prevent copying images to GitHub repository.

README.md Content

README.md file must include:

- Project title
- Student name and ID
- Brief project overview
- Dataset description
- Instructions to download the dataset
- Installation instructions
- Required Python packages
- Instructions for running the notebook
- Summary of key findings

Saving Outputs

All key visualisations should be saved to the **figures/** folder using relative paths, so they display directly on GitHub without needing the notebooks to be run. The final trained model should be saved to the **models/** folder. Both folders must be committed to the repository.

Documentation within Notebooks

Structure each notebook with clear Markdown headings. Every code block should be preceded by a Markdown cell briefly outlining what the code does and followed by a Markdown cell interpreting the results it produces. Any preprocessing or modelling decision should be justified with reference to patterns or evidence observed in the data.

All code must be clearly commented throughout.

Visualisation Standards

Every visualisation must include an informative title, labelled axes (with units where relevant), a legend where appropriate, and a short-written interpretation explaining what it reveals about the data. Files saved to the **figures/** and **models/** folders must use clear, descriptive filenames that indicate their contents at a glance.

Task 1: Dataset Exploration (15 Marks)

Before developing any deep learning models, analyse the Kahikatea image dataset to understand its structure, characteristics, and potential challenges for image classification.

Describe the dataset organisation analysing the class distribution and image properties such as dimensions, aspect ratios, colour channels, brightness, and image quality, using suitable quantitative measures and visualisations. Check for data quality issues, including duplicate or highly similar images, corrupted images, unusual dimensions, and other anomalies. Interpret the findings using clear observations supported by evidence rather than presenting visualisations without explanation. Finally, explain how the findings influenced decisions about preprocessing, augmentation, class balancing, model architecture, input size, regularisation, and evaluation.

Task 2: Data Preparation (15 Marks)

Deep learning models are highly dependent on the quality of the input data. In this task, you must prepare the dataset for model development by applying appropriate preprocessing techniques. Justify any additional preprocessing steps introduced in Task 2 that were not identified or applied in Task 1.

Discuss how the chosen preprocessing steps influence the performance of the deep learning models.

Task 3: Deep Learning Modelling and Evaluation (50 Marks)

In this task, you are expected to develop, investigate, evaluate, and select deep learning models for the Kahikatea image classification problem.

Develop at least three (3) candidate deep learning models to solve the Kahikatea image classification problem. The models should represent different architectures or modelling approaches.

For example, you may investigate:

- A custom CNN developed from scratch
- A fine-tuned ResNet model
- A fine-tuned EfficientNet model
- Another suitable state-of-the-art image classification architecture

Models that differ only in the number of neurons or layers, while otherwise using the same architecture and approach, will not normally be considered substantially different.

Your model design decisions should be justified using insights obtained from the dataset exploration conducted in Task 1 and the data preparation performed in Task 2. For each model, provide sufficient detail to enable reproduction of the experiments, including:

- Model architecture and layer configuration,
- Input dimensions,
- Activation functions,
- Optimiser and learning rate,
- Batch size,
- Number of epochs,
- Data splits,
- Regularisation techniques,
- Data augmentation methods and
- Any other relevant training settings.

Design and conduct a systematic experimental investigation to determine how selected modelling choices and hyperparameters affect the performance of your deep learning models. Your investigation should be guided by one or more clearly stated research questions or hypotheses.

Examples include:

RQ1: How does the learning rate affect the classification performance of the CNN model?

RQ2: Does data augmentation improve the generalisation performance of the transfer-learning models?

RQ3: Does fine-tuning the pre-trained layers improve?

For each model architecture, investigate at least four modelling factors or hyperparameters that are relevant to the chosen approach. Examples include learning rate, optimiser, batch size, number of training epochs, dropout rate, weight decay, data

augmentation strategies, or fine-tuning strategies. You are **not expected to test every possible combination of hyperparameters**. Instead, you should design a focused and justified set of experiments that provides meaningful evidence for selecting an effective configuration.

Using the evidence obtained from your experiments, identify the best-supported configuration for each model and train a final version of that model. For each final model, you must:

1. Present the final architecture using an appropriate architecture diagram.
2. Report the final hyperparameter values.
3. Explain how the final configuration was determined from the experimental results.
4. Train the final model using the selected configuration.
5. Evaluate the final model using the selected evaluation metrics.

Conduct a fair comparative evaluation of the final models using the same dataset partitions and evaluation procedures. The choice of evaluation metrics must be justified in relation to the characteristics of the classification problem. Your evaluation should:

1. Compare the performance of all models.
2. Identify strengths and weaknesses of each model.
3. Discuss the trade-off between model performance, model complexity, number of parameters and computational cost.
4. Justify model selection using multiple forms of evidence rather than accuracy alone.

Based on the results of the modelling, experimentation, and evaluation stages, provide a final recommendation answering the following question:

Which deep learning model is most suitable for the Kahikatea image classification problem, and why?

Task 4: Critical Reflection (10 Marks)

At the end of **DL.ipynb** file, provide a **critical reflection of approximately 500–700 words** on the entire investigation, from dataset exploration and preparation through to model development, experimentation, and evaluation.

Your discussion should demonstrate your ability to evaluate the strengths and weaknesses of your approach, interpret the significance of your findings, and identify

opportunities for improvement. Reflection should be supported by evidence from your experimental results where appropriate.

Possible discussion points include:

- strengths of your solution
- limitations of the Kahikatea dataset
- limitations of the developed models
- challenges encountered during experimentation
- computational and resource considerations
- environmental, societal and ethical implications of automated vegetation classification
- opportunities for future improvement and extensions

Marking Rubric

Criteria	Description	Marks
Dataset Exploration	**Excellent:** Comprehensive exploration of the Kahikatea dataset. Dataset structure and class distribution investigated quantitatively; image dimensions/properties examined; representative images presented; Appropriate visualisations are provided and interpreted. Findings are explicitly connected to subsequent preprocessing and modelling decisions. **(13–15)** **Satisfactory:** Most important dataset characteristics explored, with appropriate visualisations, but some areas lack depth or interpretation. Some connection between EDA findings and modelling decisions is demonstrated. **(8–12)** **Limited:** Basic or incomplete dataset exploration. Important characteristics or classification challenges are missing. Visualisations are limited, inappropriate, or largely unexplained. Findings are not meaningfully connected to later decisions. **(0–7)**	**15**
Data Preparation	**Excellent:** Appropriate and reproducible preprocessing pipeline developed. Dataset split correctly with no data leakage. Image resizing, normalisation, labelling and data augmentation are correctly implemented where appropriate. Each decision is justified using evidence from Task 1. The likely influence of preprocessing choices on model performance is discussed. **(13–15)**	**15**

	Satisfactory: Most preprocessing requirements are correctly implemented. Some justification is provided, but links to Task 1 findings or discussion of their effect on performance are incomplete. **(8–12)** **Limited:** Important preprocessing steps are missing or incorrectly implemented. Decisions are largely unjustified, or there is evidence of data leakage or inappropriate preprocessing. **(0–7)**	
Model Development	**Excellent:** At least three substantially different deep-learning models are implemented, including a custom CNN and suitable pre-trained/state-of-the-art architectures. Models are correctly designed and implemented. Architectural choices are justified using dataset characteristics and previous findings. Sufficient implementation detail is provided for reproducibility. **(9–10)** **Satisfactory:** Three models are implemented and generally appropriate, but differences between architectures, implementation details, or design justification are incomplete. **(6–8)** **Limited:** Fewer than three substantially different models, inappropriate architectures, or models that differ only through minor changes to layers/neurons. Limited justification or implementation evidence. **(0–5)**	**10**
Experimental Investigation & Hyperparameter Tuning	**Excellent:** A clear experimental investigation is designed around meaningful research questions/hypotheses. At least **four relevant factors/hyperparameters are investigated for each selected architecture**, such as learning rate, batch size, optimiser, epochs, dropout, augmentation or fine-tuning. Experiments are systematically controlled and results clearly visualised. Effects on convergence, underfitting, overfitting and generalisation are critically analysed. The best configuration for each architecture is identified using evidence rather than arbitrary selection. **(17–20)** **Satisfactory:** Four or more factors are investigated for the models, but experimental design, control, visualisation, interpretation, or justification is incomplete. Some conclusions rely primarily on performance scores. **(12–16)** **Limited:** Minimal or poorly controlled tuning. Hyperparameters are changed without clear experimental rationale, important factors are	**20**

	omitted, or results are reported without meaningful analysis. **(0–11)**	
Comparative Evaluation of Models, Final Model Selection & justification	**Excellent:** Best configurations of all candidate models are fairly evaluated using multiple appropriate classification metrics, with clear justification for metric selection. Confusion matrices and relevant error analysis are presented. Strengths and weaknesses of each model are identified. Performance, generalisation, overfitting, model complexity, number of parameters and computational cost are compared. Final architectures and hyperparameters are clearly documented. The final model is selected using strong evidence and **not accuracy alone**. **(17–20)** **Satisfactory:** Candidate models are compared using appropriate metrics, but some metric justification, error analysis, complexity comparison, or connection between experimental findings and final selection is incomplete. **(10–16)** **Limited:** Only basic aggregate metrics are reported. Little error analysis or meaningful comparison is provided. Final selection is primarily based on one metric such as accuracy. **(0–9)**	**20**
Critical Reflection	**Excellent:** Provides a critical, evidence-based reflection across the complete investigation. Strengths and limitations of the dataset, preprocessing, architectures and experiments are discussed using actual results. Computational/resource considerations are addressed. Environmental, societal and ethical implications of automated vegetation classification are considered. Meaningful future improvements/extensions are proposed. Discussion demonstrates understanding of the significance of the findings rather than simply describing the workflow. **(8–10)** **Satisfactory:** Most reflection areas are addressed, but some discussion is descriptive, lacks evidence, or is not fully connected to experimental findings. **(5–7)** **Limited:** Reflection is brief, generic, or largely disconnected from the actual experiments and results. **(0–4)**	**10**
Professional Report, Code & Repository	**Excellent:** Both notebooks execute successfully from beginning to end in a clean environment. Clear headings and Markdown explanations are provided before major code sections. Results include appropriately titled/captioned figures and tables	**10**

	with concise interpretations. Code is well organised, readable and commented where necessary, with meaningful variable names and reproducible settings such as fixed random seeds. README contains project title, student details, project overview, dataset description, installation requirements, required packages, notebook execution instructions and key findings. Figures and final models are saved using clear filenames. Repository demonstrates progressive development through meaningful commits with descriptive messages throughout the project. **(8–10)** **Satisfactory:** Repository and notebooks are generally complete and organised, but there are minor gaps in documentation, reproducibility, code quality, README content, saved outputs, or commit history. **(5–7)** **Limited:** Repository is incomplete or difficult to reproduce. Notebooks contain significant organisational/documentation problems, README is missing important information, code quality is poor, or there is minimal evidence of progressive development. **(0–4)**	
		100

References

Jia, Y. (2021). Kahikatea dataset with human-annotated true explanations for positive examples [Dataset]. In *Zenodo (CERN European Organization for Nuclear Research)*. https://doi.org/10.5281/zenodo.5059769